

CITS3002: Lab 4 - Solution

Exercise 1:

Consider a shared communication channel using ALOHA protocols where the frame time is 10 ms, there are 10 terminals, and each terminal generates frames (including retransmissions) with a mean rate of 1 frame per second.

- a) What is the probability that exactly one frame is generated during one frame time (10 ms) across all terminals?

Answer:

We model arrivals as a Poisson process. Each terminal transmits frames at the rate of 1 frame/sec. So, the total rate of 10 terminals is $\lambda=10$ frames/sec. The frame time is $T = 10 \text{ ms} = 0.01 \text{ s}$. As a result, the mean number of transmissions in one frame time is $G=\lambda T=10 \times 0.01=0.1$. The probability of generating exactly 1 frame per frame time is computed as $\Pr[1] = (Ge^{-G})/(1!) = Ge^{-G} = 0.09048$.

- b) Write a Python program to simulate and compute the throughput of both Pure ALOHA and Slotted ALOHA under the above conditions.
- c) Write a Python program to evaluate the performance of Pure ALOHA and Slotted ALOHA by varying G from 0.1 to 1.5 in increments of 0.1. Then, draw a chart that compares the throughput of Pure ALOHA and Slotted ALOHA as a function of G , with both curves displayed on the same graph.
- d) What are the maximum throughput values achievable for Pure ALOHA and Slotted ALOHA?

Answer:

Pure ALOHA = 0.18394, Slotted ALOHA = 0.36788

Exercise 2:

Consider a classic shared-medium Ethernet network using the CSMA/CD protocol. Assume that the channel is initially idle, and two terminals have a frame ready to transmit at the same time.

- a) Describe what happens when both terminals attempt to transmit. Explain the sequence of events step by step.

When both terminals have a frame ready to transmit and the channel is initially idle, each terminal first senses the channel and detects that it is free. As a result, both terminals begin transmitting immediately according to the 1-persistent CSMA protocol. However, due to propagation delay, each terminal is unaware that the other one has also started transmitting at nearly the same time, so their signals overlap on the channel, causing a collision. While transmitting, both terminals monitor the channel and detect the collision, at which point they

immediately stop frame transmission and send a jam signal to ensure all nodes recognize the collision. Afterward, both terminals apply the binary exponential backoff algorithm and wait for a random period before attempting to retransmit.

- b) After a collision, the terminals apply the binary exponential backoff algorithm. Write a Python program that simulates this process and determines after how many retransmissions each frame is successfully delivered.

Exercise 3:

Consider a classic shared-medium Ethernet network. Suppose that there are 20 stations, and each station transmits in a contention slot with a probability of 0.4.

- a) Calculate the probability that one station transmits successfully in a contention slot.

Answer:

The probability that exactly one station transmits successfully in a contention slot is:

$$A = kp(1 - p)^{k-1} = 20 \times 0.4 \times (1-0.4)^{19} = 4.87 \times 10^{-4}$$

- b) Calculate the mean number of slots per contention interval.

Answer:

The mean number of slots per contention interval is $1/A = 1/0.000487 = 2053$

- c) Assume that the frame transmission time is $200 \mu s$ and the one-way propagation delay is $\tau = 25 \mu s$. Given that each contention slot has duration 2τ , calculate the channel efficiency.

Answer:

The channel efficiency is computed as

$$\eta = (\text{TransmissionDelay}) / (\text{TransmissionDelay} + 2 \times \text{PropagationDelay}/A)$$

$$\eta = 200 / (200 + 2 \times 25 / 0.000487) = 0.00194$$